EG1113 - C Programming for Engineering Lab

Report #11

Race Preparation

Date of the experiment: November 13, 2024
Due date of report: November 15, 2021

Name: Nicholas Galus

1. Core concepts learned

- Creating a program to control a car efficiently
- Implement a feature to have an advantage

2. List of equipment and software used

- Dell Latitude
- Ubuntu
- VS Code

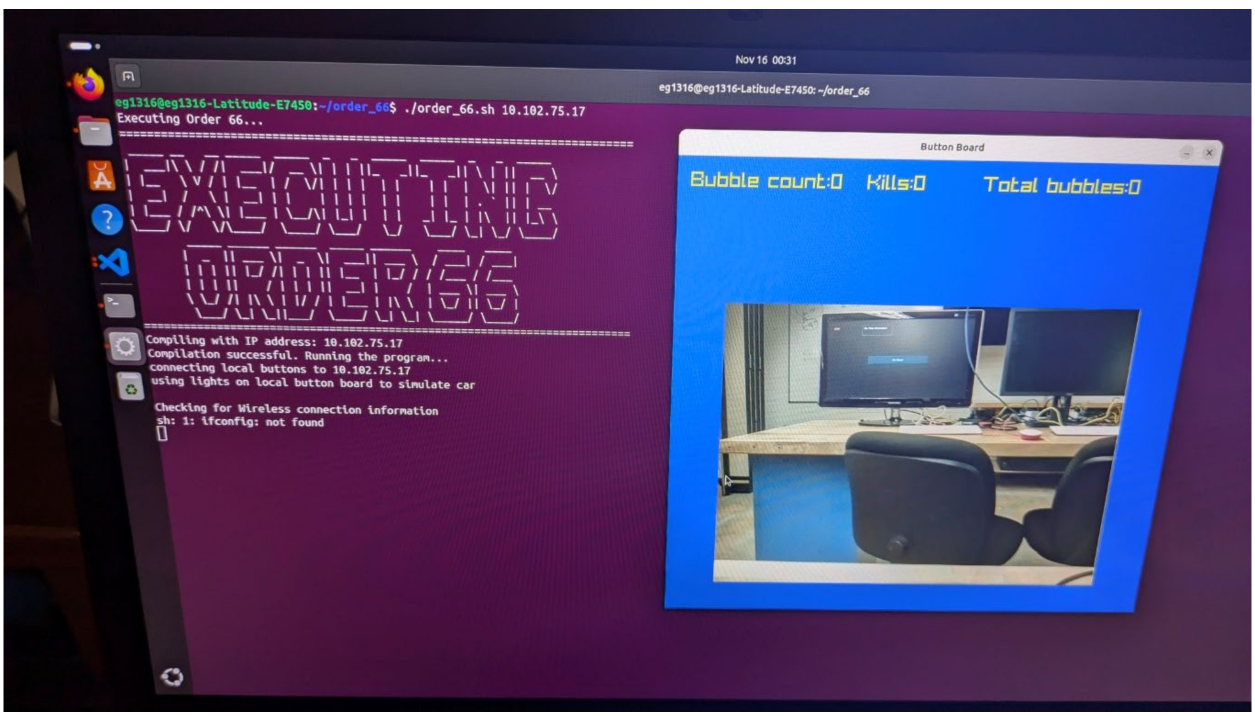
3. Relevant web links and other documents:

- None were utilized to prepare for race

4. Exercise Adding a Feature (Repeat for each exercise)

4.1 Summary

- Add a feature to program
- Remove all warnings
- Lab Verification:

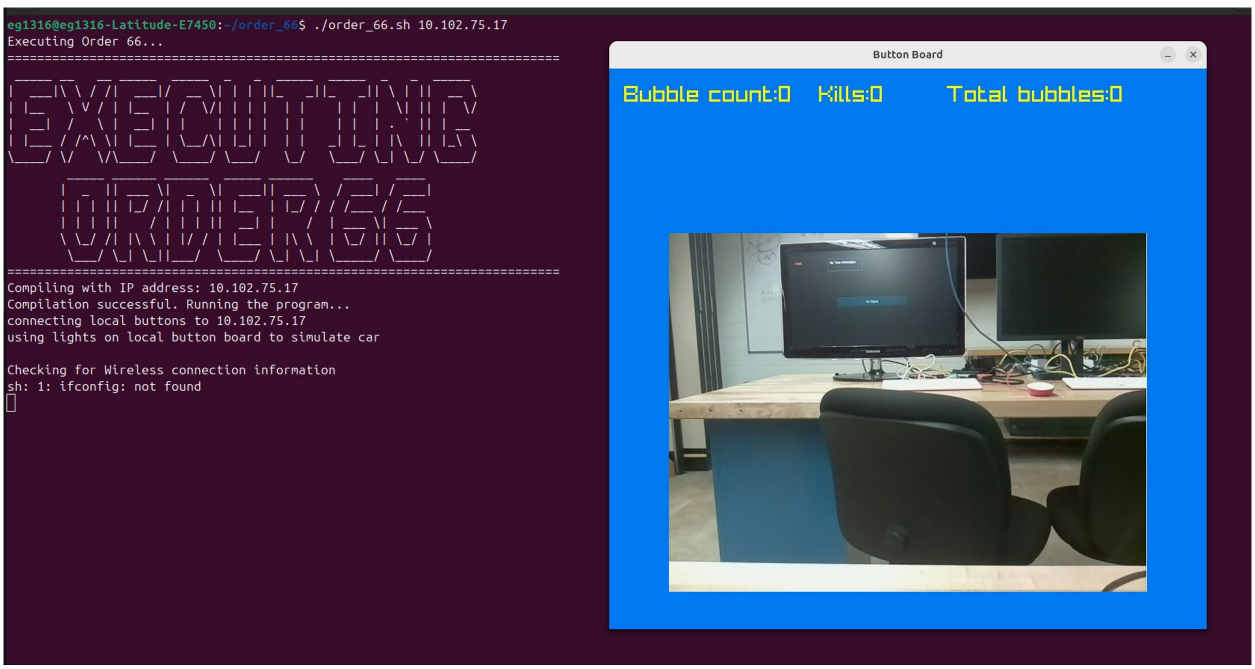


4.2 Procedures

1. Modify code to include all of the trims hardcoded into the program

```
//
//                              off     lf fw    lf bk      -       rt bk   spin lf   bk str     -       rt fw    fw str  spin rt    -        -        -        -        -
//                               0        1        2        3        4        5         6        7        8        9        10       11       12       13       14       15
int trim_percent[5][4][16]={{{   0,     100,       0,       0,       0,     100,       0,       0,       0,     100,       0,       0,       0,       0,       0,       0},
                             {   0,       0,     100,       0,       0,       0,     100,       0,       0,       0,     100,       0,       0,       0,       0,       0},
                             {   0,       0,       0,       0,     100,     100,     100,       0,       0,       0,       0,       0,       0,       0,       0,       0},
                             {   0,       0,       0,       0,       0,       0,       0,       0,     100,     100,     100,       0,       0,       0,       0,       0}},

                            {{   0,     100,       0,       0,       0,     100,       0,       0,       0,      90,       0,       0,       0,       0,       0,       0},
                             {   0,       0,     100,       0,       0,       0,      90,       0,       0,       0,     100,       0,       0,       0,       0,       0},
                             {   0,       0,       0,       0,     100,     100,     100,       0,       0,       0,       0,       0,       0,       0,       0,       0},
                             {   0,       0,       0,       0,       0,       0,       0,       0,     100,     100,     100,       0,       0,       0,       0,       0}},

                            {{   0,     100,       0,       0,       0,     100,       0,       0,       0,      85,       0,       0,       0,       0,       0,       0},
                             {   0,       0,     100,       0,       0,       0,      85,       0,       0,       0,     100,       0,       0,       0,       0,       0},
                             {   0,       0,       0,       0,     100,     100,     100,       0,       0,       0,       0,       0,       0,       0,       0,       0},
                             {   0,       0,       0,       0,       0,       0,       0,       0,     100,     100,     100,       0,       0,       0,       0,       0}},

                             {{  0,     100,       0,       0,       0,     100,       0,       0,       0,      70,       0,       0,       0,       0,       0,       0},
                             {   0,       0,     100,       0,       0,       0,      70,       0,       0,       0,     100,       0,       0,       0,       0,       0},
                             {   0,       0,       0,       0,     100,     100,     100,       0,       0,       0,       0,       0,       0,       0,       0,       0},
                             {   0,       0,       0,       0,       0,       0,       0,       0,     100,     100,     100,       0,       0,       0,       0,       0}},

                             {{  0,     100,       0,       0,       0,     100,       0,       0,       0,      95,       0,       0,       0,       0,       0,       0},
                             {   0,       0,     100,       0,       0,       0,      95,       0,       0,       0,     100,       0,       0,       0,       0,       0},
                             {   0,       0,       0,       0,     100,     100,     100,       0,       0,       0,       0,       0,       0,       0,       0,       0},
                             {   0,       0,       0,       0,       0,       0,       0,       0,     100,     100,     100,       0,       0,       0,       0,       0}}};
```

2. Remove all warnings
3. Feature: executing program utilizing a bash script



7. Problems experienced and areas of concern

- No noteworthy problems

8. Course goals

Rank the level (0=none to 5=a large amout) that this lab covered the each of the course goals by updating the appropriate box in the table below.

Course Goals	Laboratory												
	01	02	03	04	05	06	07	08	09	10	11	12	13
To compile, link, debug and run C programs	4	5	5	5	5	5	5	5	0	0	5		
To use the basic C program structure; variables, constants, and operators	4	4	4	4	4	5	5	5	0	0	5		
To use the repetition constructs such as looping with for, while, and do-while statements	2	3	4	4	4	4	4	4	0	0	5		
To use the selection structures such as if, If/else, switch, conditional expression statement	0	0	3	3	3	3	4	4	0	0	5		
To create program modules using functions (passing data to and returning values from functions)	0	0	0	2	4	4	4	4	0	0	5		
To use arrays and pointers	0	0	0	0	4	3	4	4	0	0	5		
To work with strings and string manipulating functions	2	4	4	1	2	1	2	4	0	0	5		
To perform file, I/O operations	3	4	4	5	5	5	5	5	0	0	5		
To develop structured, modular, and top-down design of software	0	0	0	1	1	4	4	4	0	0	5		